

Research papers

Thermal energy storage systems using bio-based phase change materials: A comprehensive review for building energy efficiency

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ABSTRACT

The construction industry has seen significant growth in energy consumption over the past few years, primarily due to the demand for heating and cooling in buildings. This increase in energy use presents considerable environmental and economic challenges, making reducing energy consumption in homes a critical issue for many countries. A promising approach to improving energy performance in homes while reducing CO₂ emissions is integrating phase change material (PCM)-based thermal energy storage (TES) systems into building designs. This review focuses on using bio-based phase change materials (BPCMs) in TES applications, which could contribute to lower energy consumption in the construction sector. Recent advancements in BPCM technology indicate that these substances, derived from renewable resources such as paraffin and fatty acids, can achieve thermal storage capacities similar to traditional PCMs. BPCMs function like thermal batteries, absorbing, storing, and releasing thermal energy through phase transitions, typically between 20 °C and 30 °C. This process helps stabilize indoor climates and decreases reliance on mechanical heating and cooling systems. This article comprehensively assesses the properties, thermal performance, and applications of BPCMs in building materials. It also provides an essential evaluation of life cycle assessment (LCA) and financial feasibility. Its purpose is to guide the research, engineering, and policy communities in adopting practical applications for sustainable construction practices. Integrating BPCMs in TES structures is crucial for improving sustainability and energy conservation in the construction industry. This approach offers a viable pathway toward creating more energy-efficient buildings. Furthermore, a comparative analysis is provided, evaluating BPCMs along different varieties of section trade substances, inclusive of natural, inorganic, and eutectic alternatives. This analysis specializes in numerous crucial factors: price, energy performance, and environmental impact. The review addresses challenges such as heat transfer costs, compatibility with conventional building materials, and policy implications.

1. Introduction

The requirement for energy is more significant than it has ever been due to these rapidly growing economies. The limited supply of fossil fuels prevents them from meeting this demand [1]. Solar energy is among the most affordable and widely available forms of renewable energy. However, the sporadic nature of renewable energy sources, such as solar energy, poses a severe drawback. The capacity to retain thermal energy so becomes crucial [2]. A medium specifically storing energy can store the extra thermal energy generated during the day. Cost, storage density, environmental impact, reliability, and other considerations are crucial when selecting a material for heat storage. As phase change materials (PCMs) demonstrated the capacity to collect and release

thermal energy during a material's phase transition, there is now an increased interest in researching the potential uses of these materials [3]. However, the drawbacks of these materials are their low heat conductivity and the problem of liquid leakage. Leaks, for instance, must be prevented by adequately managing the transition from solid to liquid. Enhancing the performance and stability of pure PCMs may be achieved through two distinct methods [4]. PCMs' low heat conductivity is another technological challenge that has to be solved. The heating and lighting of homes consume more than 40 % of the energy generated worldwide. This amounts to 33 % of all greenhouse gases emitted into the environment annually.

Energy storage technologies are essential to the energy industry, particularly when it comes to boosting building energy efficiency and

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integrating renewable energy sources. These systems can improve energy efficiency by up to 30 %, depending on the specific technology and consumption patterns. For example, lithium-ion batteries, which have an energy density of approximately 150–250 Wh/kg, allow for efficient energy storage and retrieval. The levelized cost of energy storage for lithium-ion batteries has decreased to around \$150 per MWh, making them economically viable for both residential and commercial applications. In comparison, pumped hydro storage typically ranges from \$100 to \$200 per MWh, depending on geographical and operational factors. Furthermore, energy storage systems provide peak shaving capabilities, which can reduce peak demand in buildings by 20–40 %, leading to significant savings on energy bills. When combined with renewable energy sources, energy storage systems can lower greenhouse gas emissions by about 30 %, a crucial benefit in high-consumption urban areas. Life cycle assessments of lithium-ion batteries reveal a cycle life of 2000 to 5000 cycles, translating to a lifespan of 5 to 15 years, depending on usage conditions and management strategies. Additionally, TES systems in buildings can depot energy equivalent to 10–20 % of daily energy consumption, effectively shifting consumption from peak to off-peak hours [5].

Better control over energy usage in residential buildings is required. Innovative techniques must manage temperature fluctuations and maintain internal temperature using eco-friendly methods. This may be carried out by and large thru thermal energy storage (TES), in particular thru latent heat energy storage (LHES) in bio-based phase change materials (BPCMs). BPCMs possess specific chemical, physical and thermal characteristics, making them essential for meeting energy management specifications. Utilizing PCMs allows constructing additives to correctly store thermal strength, mitigating the influences of energy fluctuations. They are employed in various programs, consisting of air conditioning and solar heating structures. When used together with substances like concrete and gypsum, PCMs enhance the sustainability of structures [5,6]. Wood and timber fibers also are suited for production because of their reliable properties and aesthetic attraction [7].

Addressing energy and ecological demanding situations requires a focal point on energy production and storage systems that increase energy security within building clusters via utilizing renewable energy resources (RES) [8]. The building cluster method optimizes energy control and overall performance. A complete framework for urban energy structures has been proposed, emphasizing the need to manipulate energy consumption and generation collectively in place of in isolation [9,10]. Research on renewable energy optimization and energy overall performance on the community scale highlights the importance of defining barriers for constructing clusters. This introduces the time period “multi-building energy system” (MBES) to unify various interpretations and make clear relationships among homes, the energy grid, and energy storage. Understanding these interactions can cause progressed energy distribution and performance in urban settings. Morphology and density analysis might also similarly enhance energy simulation accuracy and reduce emissions [11].

This article focuses on analyzing the integration of PCMs and the challenges of incorporating PCMs into construction materials. The research findings revealed several interesting areas that require more study, such as developing new PCM materials with improved thermal characteristics, improving PCM-based cooling systems, and using PCM in construction materials and buildings. Several BPCM types may be directly applied to lower building energy consumption. The scientists identified many deficiencies in BPCMs found during the encapsulation process, such as low phase change latent heat, poor thermal conductivity, and liquid leakage. Additionally, the preparation procedure becomes more complex and expensive when utilizing additional micro-encapsulation technologies. The most mature technology is based on microencapsulating PCM into a polymeric matrix; however, this method is not easy, requires more expensive reactive materials to prepare the polymeric matrix, and supposes a reduction in the heat transfer process. Energy-efficient bituminous building materials have been prepared as

an alternative, including PCM and a polymer (like styrene-butadiene-styrene). Thus, SBS can create a polymer matrix that prevents PCM leakage. In 2022, Wang et al. [12] conducted a comprehensive examination of BPCMs and their processing methods, dividing BPCMs into two categories: plant-based porous materials and animal-based porous materials. This field of study has examined activation, hydrothermal carbonization, and direct carbonization as processing methods.

BPCMs are being utilized in various packages, which include industrial waste heat recovery, solar energy storage, and constructing thermal management. They offer capacity financial and environmental benefits via enhancing thermal energy storage and average profitability. In 2023, Mehrizi et al. [13] centered on BPCM applications for efficient heat control across incorporated structures. Research on thermal energy storage technology targets to deal with the intermittent nature of energy sources [14]. PCMs are commonly used for this motive, as they are able to take in or release thermal energy in the course of phase transitions [15,16]. PCMs increase a construction's thermal inertia and energy performance due to their excessive energy density and storage ability [17–20]. They can also help shift energy demand to off-height hours, making them treasured in constructing thermal control. While there has been considerable research on solar heating, the integration of PCMs with solar thermal systems remains limited, highlighting a multidisciplinary area for exploration. Topics include the use of PCM nanofluids in solar collectors and applications of thermal storage units for building heating [21–23]. Biao Liu and colleagues [24,25] analyzed the thermal performance of PCM/solar heating systems but noted a lack of comprehensive development in current frameworks. More research is needed on integrating solar thermal systems with thermal energy storage for building heating, especially in comparing active and passive approaches.

1.1. Methodology involved

This study systematically reviews articles on thermal energy storage systems that utilize BPCMs in improving building energy efficiency. The topics are limited to bio-based phase change materials and their utilization in thermal energy storage systems with respect to the building energy efficiency, which will be used as the selection criteria. The method we use to collect data in our research is literature search in some common scientific databases which are Scopus, Web of Science and Science Direct. These three can be considered as the largest databases that have the most academic and scientific literatures which are suitable for us to get data and information necessary for our research. To maximize the search results, a combination of specific keywords is employed. These keywords are pivotal in uncovering relevant information across various research domains. Typical keywords include “bio-based phase change materials,” essential for innovative thermal energy storage solutions. Another key phrase is “thermal energy storage” itself, highlighting technologies that contribute significantly to building energy efficiency. The term “sustainable construction” is also crucial, reflecting the growing emphasis on environmentally friendly building practices. Lastly, “PCM integration in buildings” is a critical keyword, denoting the incorporation of PCMs to enhance the performance and sustainability of building designs.

The exclusion criteria for the selection of articles include several vital considerations. The articles that exclusively focus on non-bio-based PCMs are excluded from consideration. Additionally, studies that fail to provide empirical evidence or relevant data will not be included. Papers that have not undergone the peer-review process or are published in journals that are not considered reputable are also excluded. Critical information, including authors, publication year, methodology, findings, and conclusions, is meticulously extracted from selected articles in data extraction and analysis. Following this, each study's quality is assessed, considering factors such as the methodology employed, the sample size, and the robustness of the results. Subsequently, the extracted data is synthesized to identify trends, gaps, and future research

directions in bio-based PCMs for thermal energy storage in construction. This systematic approach ensures that the review is comprehensive, relevant, and scientifically rigorous, providing valuable insights into integrating bio-based phase change materials in enhancing building energy efficiency.

2. Energy efficiency in buildings

2.1. Energy usage and thermal comfort in buildings

Over the past ten years, the need to enhance thermal comfort in the built environment has increased energy use. From 2010 to 2023, this demand expanded by around 7.03 %, from 119 k (EJ) to 128 k (EJ). The primary sources of building energy uptake are HVAC (Heating, Ventilation, and Air Conditioning) systems, which are crucial to providing an inviting internal environment comprising around 38 % of the building's total energy use. The natural world, the economy, and society are all challenged differently by buildings' escalating energy usage. Local energy supplies may be strained as a result, which might result in outages and higher utility expenses.

It additionally accelerates long-term climate change by releasing more greenhouse gases. Approximately 40 % of greenhouse gas emissions come from the building industry [26]. Prioritizing energy-efficient building designs, encouraging renewable energy sources, and implementing sustainable behaviors are essential to addressing these ramifications. Based on the temperature differential between the interior and outside environments, structure thermal efficiency significantly impacts how much energy a structure uses. The building exterior is essential to attaining energy-efficient building performance [27]. To meet the thermal needs of the buildings, it is necessary to manage the loads of heating and cooling in interior and outdoor spaces [28]. To save energy, researchers have employed a variety of advances to boost the thermal efficiency of the building coverings. Integrating thermal energy storage devices into the building envelope is one of the effective tactics that has demonstrated encouraging results.

2.2. Thermal energy storage technologies

Thermal energy storage is a very effective and assuring technology, and it is anticipated to significantly impact the optimization and regulation of thermal energy usage. It permits transient thermal energy storage due to a material's internal energy changing when its temperature rises [29]. When needed for cooling and heating, this stored energy might be used later [30,31]. Industrial operations and the building industry greatly benefit from using TES systems. Using thermal energy storage technology in building construction can significantly improve overall energy efficiency. These technologies significantly lower total energy consumption because they may narrow the energy gap between provision and requirement [32]. The three primary methods for efficiently storing thermal energy are sensible heat, latent heat, and thermochemical processes. The applications for which these techniques are employed are determined by the benefits and drawbacks each offers [33]. Thermochemical techniques contain storing energy through reversible chemical reactions. In those procedures, energy is absorbed or released as the chemical bonds are broken and reformed, taking into account high energy density storage. This approach is mainly tremendous due to its capacity to store energy for prolonged periods without sizable losses and its ability to offer lengthy-time period storage solutions. Thermal energy storage for homes offers both qualitative and quantitative advantages. Qualitatively, it enhances indoor comfort via preserving steady temperatures, reducing dependency on outside energy resources, and lowering emissions. Quantitatively, it could cause extensive energy financial savings, with research showing discounts in energy consumption through up to 30 % in homes prepared with efficient TES systems. One instance of a TES system in buildings is the usage of PCMs in partitions or ceilings, which imbibe and release heat to

maintain stable indoor temperatures. Another example is the setup of water tanks for storing hot water generated for the duration of off-peak hours for later use, lowering the need for continuous heating. Additionally, ice storage systems may be used in commercial homes to provide and store ice at night, which is then used for cooling purposes all through the day. Future advancements in thermal energy storage technology are likely to focus on improving the efficiency and cost-effectiveness of existing methods. Innovations in materials science could lead to the evolution of new PCMs with higher energy storage capacities and faster response times. Advancements in thermochemical storage could result in more sustainable and scalable solutions, potentially integrating renewable energy sources for even greater environmental benefits. In residential buildings, TES systems can be integrated into heating, ventilation, and air conditioning systems to optimize energy use. For instance, homeowners can utilize solar thermal collectors to store energy during sunny periods, which can then be used to heat water or provide space heating during cooler times (Fig. 1). Furthermore, incorporating advanced PCMs into residential construction materials can greatly enhance energy efficiency by steadying indoor temperatures, repressing reliance on traditional heating and cooling systems, and lowering utility bills.

Sensible heat storage: Thermal energy can be saved for later release by reducing the temperature of a solid or liquid after it has been raised without resulting in a phase change [34]. The quantity of thermal energy stored may be calculated based on the mass, temperature differential, and specific heat capacity of the storage medium (Fig. 2(a)). When a material changes from a solid to a liquid or vice versa, latent heat storage is the release or preservation of heat during that process [35]. Construction applications frequently employ solid-liquid phase transition because it provides almost isothermal heat storage and recovery and a high energy storage density. As a substantial amount of sensible and latent heat is melted, the PCM retains it (Fig. 2(b)). The thermochemical storage of heat is an additional technique for storing thermal energy. This mechanism stores and releases thermal energy through a completely reversible chemical reaction resulting from the breaking and re-forming of molecular bonds. An endothermic reaction occurs when the thermochemical substance (AB) is heated to the point where it separates into components A and B. A and B, the two separate components, retain heat during this operation. The thermal energy is released when A and B recombine due to an exothermic process (Fig. 2(c)). Scientists and engineers are turning to thermochemical heat storage more frequently to find new uses for thermal energy that can be stored, such as waste heat recovery, the integration of renewable energy sources, and industrial processes [36,37].

2.3. Latent heat storage and PCMs

Latent heat storage differs from the other thermal energy storage techniques previously addressed in that it can store heat at a temperature that is almost constant and consistent with the phase change material's phase transition temperature. Moreover, it offers a high density of energy storage. PCMs are substances that, depending on their physical condition, have the capacity to either emit or absorb large amounts of latent heat [38]. Because PCMs have a high heat capacity within a narrow temperature range, they may be used in nearly isothermal heat storage. These substances have the ability to change into liquid-gas, solid-liquid, and solid-solid [39,40]. PCMs are primarily classified as inorganic, eutectic, and organic compounds [41]. It is categorized further based on the chemical constituents of the PCM. A thorough schematic illustration of these grouped PCMs is shown in Fig. 3. Organic PCMs come into two categories: paraffinic and non-paraffinic materials, the latter of which includes subcategories such as alcohols, fatty acids, esters, and glycols. Organic PCMs are mostly composed of carbon-hydrogen links. Applications for heat storage, particularly at lower to medium temperatures, frequently make use of these organic PCMs [42]. High storage density, a suitable phase transition temperature range of

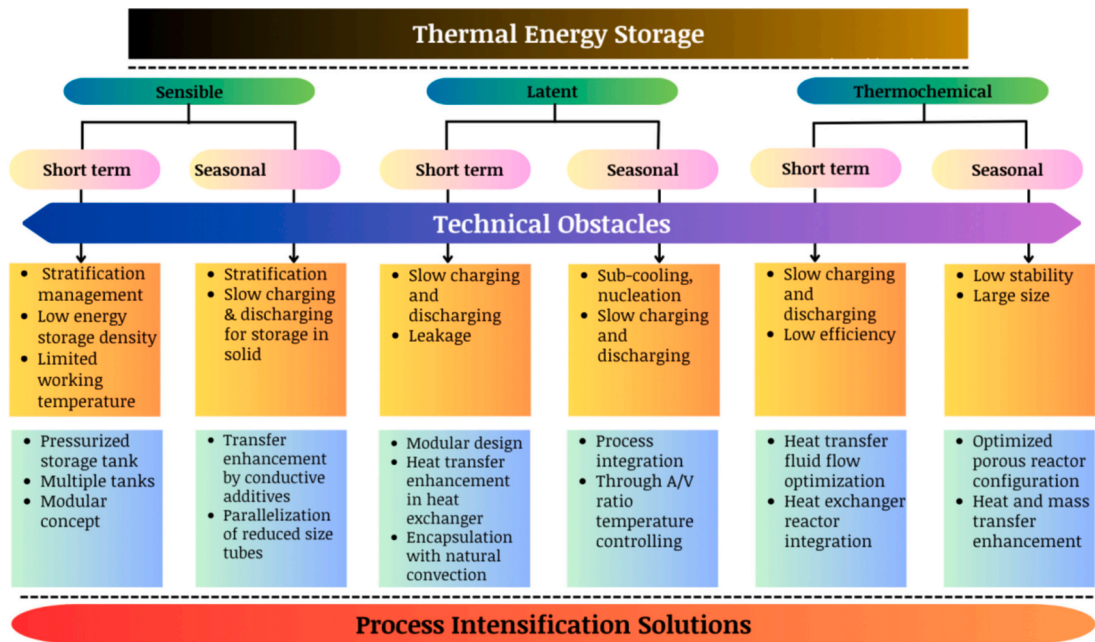


Fig. 1. Overview of thermal energy storage systems.

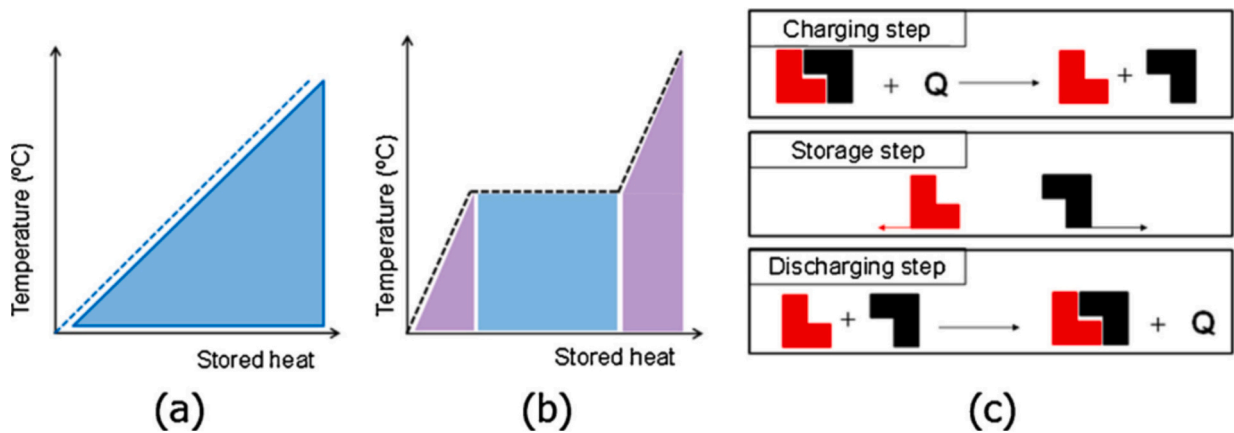


Fig. 2. TES methods, (a) Sensible heat, (b) Latent heat, (c) Thermochemical reaction. Reproduced from Ref. [37]. Copyright (2023), with permission from Elsevier.

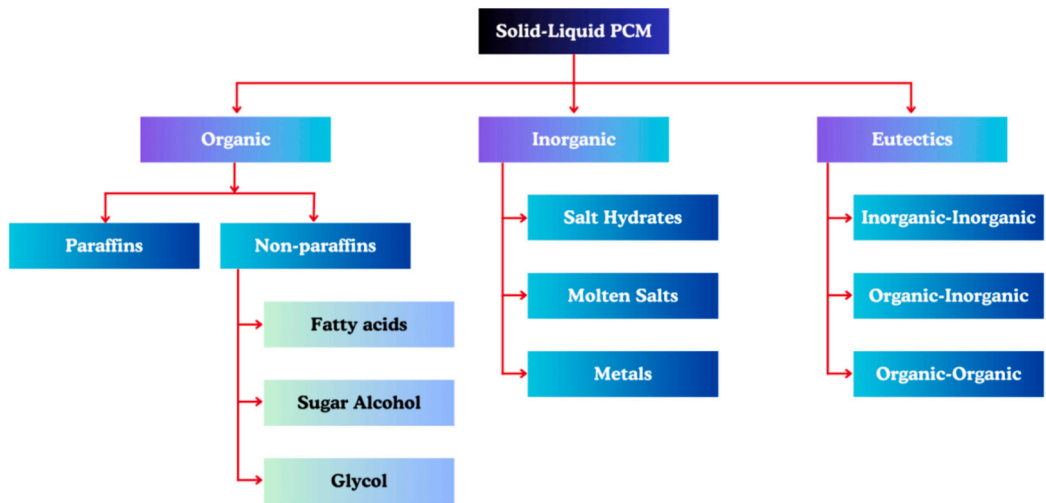


Fig. 3. Classification of phase change materials.

0–150 °C, and a stable chemical and physical composition are some advantages [43].

In contrast, materials such as metals and salts make up inorganic PCMs, which lack carbon or hydrogen atoms. Their usefulness for solar applications stems from their ability to melt at temperatures between 100 °C and 1000 °C, which are about ten times greater than organic PCMs. When two or more PCM types are combined, a crystalline mixture is formed during the crystallization phases. This is made possible by eutectic PCMs, which enable each component's congruent melting and solidification. Compounds that are organic-inorganic, organic-inorganic, or inorganic-inorganic can form eutectic combinations. Different materials can be mixed to adjust latent heat, melting points, and thermal conductivity to meet specific requirements.

Although little is known about the characteristics of eutectic PCM, some notable benefits are as follows: an acceptable volumetric heat storage density and well-defined melting temperatures. Each kind of PCM—eutectic, inorganic, or organic—must have specific benefits and drawbacks to provide the best thermal efficiency for building applications. This emphasizes how crucial it is to give these characteristics careful consideration. The advantages and disadvantages of every kind of PCM are presented in Table 1. The sampling principle governs PCM usage. PCMs go through an endothermic process that turns them from a solid into a liquid when the temperature rises, and the melting point is achieved. There is a steady temperature during this operation. However, PCM releases heat by an exothermic mechanism and resolidifies when the temperature drops below the melting point [44]. PCMs can save 5–14 times as much thermal energy as their traditional counterparts in a given unit volume. Aridi and Yehya compared the storable energy of several materials, including PCM, wood, plaster, concrete, iron, glass wool, and water [45]. PCM has a storable energy of 287.6 Wh/m², compared to 92.8 Wh/m² for water. To lower energy consumption and

increase thermal efficiency, selecting the right PCMs to add to building envelopes is a crucial step that influences several factors [46]. These requirements include volume change during phase transition, congruent melting, latent heat, melting point, specific heat capacity, vapor pressure at working temperatures, and thermal conductivity at operating temperatures.

Qualitatively, the choice of PCM can greatly influence building performance by providing stable indoor temperatures and reducing energy demands, thanks to their capacity to captivate and release heat efficiently. The qualitative blessings also enlarge to enhancing occupant consolation and minimizing HVAC gadget stress, making them a treasured asset in sustainable building design. PCMs are also cost-energyful and can be hooked up with minimal disruption. Moreover, they're non-toxic and biodegradable, making them an eco-friendly choice. Compared to other energy storage solutions, together with batteries or flywheels, PCMs offer a greater passive and maintenance-free approach to thermal management. While batteries can save and release electrical strength, PCMs attention specially on thermal energy, making them ideal for applications in which temperature law is prime. Additionally, not like mechanical solutions like flywheels, PCMs do not require moving elements, reducing the chance of mechanical failure and preservation charges. Despite their advantages, implementing PCMs in building designs can present several demanding situations. One essential trouble is the preliminary price of PCM materials, which can be higher than conventional constructing substances, significantly affecting project budgets. Additionally, right integration into current structures requires careful attention of fabric compatibility and specific set up strategies to ensure most efficient overall performance and avoid problems such as thermal bridging or material degradation through the years. To deal with the high preliminary value of PCM substances, bulk purchasing agreements or government incentives can be explored to lessen prices. Additionally, developing standardized installation strategies and training applications for contractors can make certain right integration and decrease errors. Research into advanced PCM formulations with progressed cloth compatibility also can help in overcoming challenges associated with thermal bridging and degradation.

The kinetic characteristics highlight high nucleation and crystallization costs, which are likewise noteworthy. Chemical stability, reversible melt/freeze cycles, compatibility with production materials, and safety considerations are other attributes that ought to be taken into consideration. Economic factors become even more important as efficiency and wide distribution become critical. Specifically, it is difficult for anyone PCM to fully satisfy these requirements. Because of the significant volume change during a phase shift, PCMs are considered useless for most applications. The food industry, solar water heaters, photovoltaic cells, apparel and textiles, the automobile industry, electronic devices, and thermal regulation in buildings are just a few industries that use PCMs [47–53]. The intriguing approach of integrating PCMs into building structures is examined in this work, which is primarily focused on building applications. By effectively regulating building energy use, this approach has shown notable improvements in reducing heating and cooling system requirements. Because of their favorable qualities, such as minimal volume fluctuations and tiny pressure changes, PCMs based on solid-liquid transitions are frequently employed in construction applications [54]. Properties commonly highlighted in real-world circumstances are thermal conductivity, phase change temperature, and latent heat capacity. The building envelope systems' requirements are met by carefully adjusting these crucial qualities to achieve optimal thermal performance.

3. An overview of thermal energy storage systems and solar heating

Research suggests that the amount of solar energy received in one hour matches the global energy demand. Additionally, Earth receives over three times the global energy requirement yearly from solar

Table 1
Advantages and drawbacks of the various types of PCMs.

PCMs types	Advantages	Drawbacks
Inorganic PCMs		
Salt Hydrates	- High latent heat per unit volume - Low cost - Non-flammable	- Phase separation over time - Corrosive to metals without proper containment - Tendency to supercool
Metallic	- Extremely high thermal conductivity - High latent heat	- High cost - Difficult to handle - High density, leading to heavy systems
Eutectics	- Sharp and precise MP/FP - Can be tailored to specific temperatures by mixing different materials	- Risk of phase separation during cycling if poorly designed - Expensive than single component PCMs
Bio-PCMs	- Non-toxic - Renewable & environmentally friendly - High latent heat - Low flammability	- Limited thermal conductivity - Higher susceptibility to biodegradation if not properly encapsulated
Hydrated Salts	- High energy storage density - Easily available - Inexpensive	- Expensive - Corrosive - Prone to super cooling - Inconsistent thermal properties due to phase separation
Organic PCMs		
Paraffin	- High latent heat of capacity - Chemically stable - Non-corrosive - Wide range of MPs - Recyclable - Sharp MP/FP - Non-corrosive & non-toxic	- Low thermal conductivity - High volume change during phase transition - Flammable - High cost compared to paraffin
Fatty Acids	Good thermal stability	Slightly lower latent heat than paraffin

radiation. Solar energy is widely used for energy generation, central and district heating, and heating homes and businesses. The abundance of solar energy on Earth makes it an attractive option. However, solar energy has limitations due to its intermittent nature, which restricts its applications and leads to inefficient utilization. There is significant interest in developing thermal energy storage technologies to address the intermittency of solar energy [55]. This section will also discuss the current progress in integrating phase change materials with solar energy systems.

3.1. Configuration of solar heating system

Solar energy systems gather solar radiation and convert it to thermal or electrical energy. This study is devoted to solar heating systems, specifically the most advanced PCM solar heating system available. Photovoltaic or solar energy-producing systems are not taken into account in this study. Using forced and natural circulation as a basis for classification, solar energy systems fall into two categories [56]. The natural circulation solar energy system operates on the principle of the thermosyphon phenomena. The sun heating terminal, solar distribution system, and solar collector are the three primary parts of a solar energy system. Converting solar radiation into thermal energy is the function of several types of solar collectors [57].

3.1.1. Solar collector

Solar collectors are essential in thermal energy storage systems, particularly when integrated with BPCMs, to enhance building energy efficiency. Incorporating PCMs into solar thermal systems allows for the practical storage of thermal energy, which can be utilized during periods of low solar radiation, thereby improving the overall efficiency of energy use in buildings. One significant advancement in this area is the integration of nano-encapsulated PCMs within solar thermal collectors. This innovative approach addresses common issues such as PCM leakage and enhances the thermal storage capacity of the system.

Above other solar collector types, flat-plate solar collectors (FP-NCs) are the most widely used due to their straightforward construction. A translucent cover that allows sunlight to pass through the collector's top and covers the heat-absorbing plate that is there to absorb solar radiation. Due to its more significant heat loss, unglazed solar collectors (UG-NC), to which two manifolds are typically attached, are primarily used in low-temperature applications. Furthermore, to maximize their ability to absorb solar thermal energy, collectors are often constructed from highly thermally resistant materials and have rich colors painted on their exteriors. Considered one of the most famous collectors in the world, the evacuated tube collector (ET-NC) is utilized extensively. Its current extensive use results from its ongoing monitoring of sunlight [58]. An evacuated tube solar collector's basic construction. It is clear that they are covered with transparent glazing, and one side has a manifold for collecting and spreading thermal medium. On the other side is a frame for mounting the solar collectors, which are attached to a reflector placed underneath them to enhance the solar radiation reflected to the lucifugous portion. Although air has a far lower thermal conductivity and capacity than water in solar collectors, it is a typical hot medium for solar thermal systems.

Furthermore, the CSCs are depicted below. Typically, the reflecting dish can rotate uniaxially or even biaxially in the sun's direction. The parabolic dish reflector's focal point is precisely where the thermal receiver is placed. It collects reflected solar radiation and transforms it into thermal energy that users may use. Using nano-coated PCMs, researchers have demonstrated improved energy efficiency and reduced carbon emissions associated with solar collector systems [59]. The encapsulation of PCMs stabilizes their performance and allows for a more compact design, which is particularly beneficial in urban settings [60]. A systematic review of transpired solar collectors found that these systems can achieve up to 80 % efficiencies by utilizing PCMs for thermal storage. This high efficiency is attributed to the ability of PCMs to

absorb and release heat at specific temperatures, which aligns well with the thermal demands of buildings [61]. Using PCMs in transpired solar collectors allows for more consistent thermal output, mitigating the fluctuations caused by the intermittent nature of solar energy [62].

Moreover, studies have shown that the performance of solar collectors can be significantly enhanced by optimizing the design and materials used in conjunction with PCMs. For instance, a numerical study on U-shaped flat plate solar collectors revealed that integrating PCMs can stabilize the output heat, making it more suitable for domestic hot water systems and space heating [63]. The thermal performance of these systems is further improved by selecting appropriate materials that maximize heat absorption and minimize heat losses. Experimental investigations have also highlighted the benefits of using flat plate solar collectors integrated with PCMs. These systems have been shown to effectively reduce heat loss and improve energy savings in tropical climates, demonstrating their versatility and effectiveness in various environmental conditions [64].

Furthermore, as phase change materials make it possible to use renewable energy sources more effectively, their usage in solar thermal systems has been associated with a decrease in greenhouse gas emissions. Adding bio-based phase change materials to solar collector systems is a potential way to increase building energy efficiency. By enhancing thermal energy storage capacities and stabilizing thermal output, these systems may considerably lessen dependency on fossil fuels and promote more environmentally friendly energy practices.

3.1.2. Heating distributing system

Before discussing the heat distribution system, we talk about the solar collectors. The heating system's distribution method can be installed above, in the duct, or immediately underground, as is widely known. In a thermal medium temperature way, it may also be divided into standard and high temperatures using various shell materials, such as metal, fiberglass, or high-density polyethylene. The distribution network may be divided into heat delivery networks for steam and hot water based on internal pressure. Regarding their relative locations and interactions, there are primary and minor networks. Given the current state of the environment and resource availability, traditional fossil fuel-energized heating systems need to be upgraded to satisfy the demands of the energy landscape. Utilizing solar radiation as an auxiliary heat source may lower energy consumption and increase the heating or energy supply system's adaptability to weather conditions.

Heating distribution systems incorporating thermal energy storage using bio-based PCMs are increasingly recognized for their potential to enhance building energy efficiency. By using PCMs' unique ability to absorb and release thermal energy during phase transitions, these systems can efficiently control heating loads and lessen their need for traditional energy sources. Solar thermal systems are one prominent application for using TES with bio-based PCMs. In order to store extra thermal energy produced during peak sunshine hours for usage during low solar availability, these systems use PCMs' latent heat storage capability. Shukla and Kishan draw attention to the fact that adding PCMs to solar thermal energy storage systems may significantly enhance the efficiency of space heating applications, especially in residential structures where winter heating demands are considerable [65].

The ability of PCMs to maintain a nearly constant temperature during the phase change process allows for more stable indoor temperatures, enhancing occupant comfort while minimizing energy consumption [65]. In addition to solar thermal applications, TES systems using bio-based PCMs can also be employed in district heating networks. Offering flexibility in heat distribution, including decentralized thermal storage in these networks, can improve overall energy efficiency. Thermal accumulators are essential for maximizing energy and energy efficiency, according to Schuchardt [66], especially in poorly maintained areas of district heating systems. These systems can boost the efficiency of the heating network by reducing heat losses and releasing thermal energy stored during off-peak hours.

Moreover, using bio-based PCMs in TES systems can contribute to sustainability goals by utilizing renewable materials that have a lower environmental impact than traditional synthetic PCMs. Oluah et al. discuss the potential of nano-enhanced bio-based PCMs for passive cooling applications in buildings, which can also be adapted for heating purposes [67]. Incorporating such materials enhances thermal performance and aligns with green building practices aimed at reducing carbon emissions and energy consumption. Another innovative application uses microencapsulated phase change materials (MPCMs) in combined heat and energy (CHP) systems. Guo et al. demonstrate that MPCMs can improve the load regulation capabilities of CHP units, allowing for more efficient energy management [68]. It is feasible to enhance thermal storage efficiency and maximize energy production under variable demand circumstances by including MPCMs in the heat transfer fluids of these systems. To sum up, heating distribution systems that combine phase change materials derived from biotechnology with thermal energy storage present a viable way to improve the energy efficiency of buildings. Their application in combined heat and energy systems, district heating networks, and solar thermal systems demonstrates their adaptability and capacity to support sustainable energy practices.

3.1.3. Heating terminals

Heating terminals are a crucial component of heating systems; they primarily employ convection and radiation to transfer thermal energy to interior spaces. It is possible to have air heating equipment (AH), radiant heating devices (RH), and radiators (RA). Steam or hot water is the primary thermal medium radiator to heat the interior space through convection. There are three different kinds of radiant heating devices: low, medium, and high. Radiant heating systems use radiation to warm their surroundings. Using forced convection, the AH gadget raises the room's temperature by directly providing hot air to the space.

Heating terminals for thermal energy storage systems that incorporate BPCMs play a crucial role in enhancing building energy efficiency by utilizing the latent heat storage capabilities of PCMs to manage thermal energy effectively. This allows buildings to maintain comfortable temperatures while minimizing energy consumption. For instance, in radiant floor heating systems, bio-based PCMs can be integrated into the flooring material to absorb excess heat during the day when solar gain is high and release it during the cooler night hours, not only stabilizing indoor temperatures but also reducing reliance on conventional heating systems and leading to significant energy savings. Similarly, bio-based PCMs can be embedded in wall panels or ceiling tiles where they absorb heat during the day from sunlight and indoor activities to prevent overheating and then release stored heat at night to maintain a comfortable indoor climate without the need for additional heating, which is especially beneficial in buildings with large glass facades [69,70]. Furthermore, heating terminals can be designed to include tanks filled with bio-based PCMs, storing thermal energy generated from renewable sources like solar thermal collectors or biomass boilers and distributing it through heating terminals when needed, effectively decoupling energy generation from consumption and optimizing energy use. By storing excess thermal energy produced during off-peak hours and releasing it during peak demand periods, bio-based PCMs can increase energy efficiency and reduce operating costs in hybrid heating, ventilation, and air conditioning (HVAC) systems. Furthermore, bio-based PCMs can be added to free cooling systems to release heat at night and absorb it during the day, therefore minimizing the need for mechanical cooling in areas with considerable daytime and nighttime temperature swings.

Lauryl Alcohol/Expanded Graphite Composites and their integration into building materials significantly enhance thermal regulation within buildings. Using lauryl alcohol as a PCM and expanded graphite to improve thermal conductivity, these composites can be effectively integrated into ceramic tiles or wall panels. This integration facilitates efficient thermal storage and release, enhancing indoor comfort and minimizing energy consumption [71,72]. Additionally, PCM-Enhanced

Radiant Heating Systems leverage bio-based PCMs to store thermal energy within floor structures. During the heating season, these systems absorb heat from the hydronic system and release it gradually, ensuring the floor temperature remains consistent and comfortable, significantly reducing the energy demand for heating. Buildings become more energy efficient when bio-based PCMs, such as insulation materials, are included in the building envelope. These PCMs retain a suitable interior temperature while using less HVAC equipment by absorbing heat throughout the day and releasing it at night. Along with improving comfort, this method helps buildings be designed with more energy efficiency [73].

3.2. Various PCMs used in a solar heating system

More and more materials are included in solar heating systems to improve thermal energy storage and efficiency. PCMs are perfect for applications where controlling temperature is essential since they store and release thermal energy as a substance changes from a solid to a liquid. By storing extra solar energy during the day and releasing it when needed, PCMs significantly improve the efficiency of solar household hot water systems. Sodium thiosulfate pentahydrate is an example of a PCM that effectively maintains water temperatures for a more consistent hot water supply. In air heating, PCMs are incorporated into the thermal storage units of solar air heaters, enabling quick heating of air during sunny periods and gradual release later, ensuring a steady flow of warm air even without solar radiation. A notable application is in closed collector-storage solar air heating systems where PCMs enhance temperature retention and efficiency [74]. PCMs embedded in building materials, including floors and walls, also assist in the control of indoor temperatures by collecting excess heat during the day and releasing it at night, reducing the demand for heating and cooling systems and improving energy savings [75]. When combined with conventional solar thermal systems, PCMs significantly boost performance. Hybrid systems work alongside traditional water storage tanks to extend heat availability, storing heat during peak sunlight hours for later use, such as preheating water in a conventional tank, thereby improving efficiency. On a larger scale, in concentrated solar energy (CSP) plants, PCMs store thermal energy generated throughout the day for energy production during low solar availability periods, with molten salts commonly used due to their high thermal capacity and suitable melting points [76].

As we look at the parts of the sun-assisted heating system, thermal energy storage technology is also essential to solar heating [77]. This is because it increases the system's adaptability to inclement weather and flexibility in providing heating loads. Systems for storing thermal energy can be classified as either sensible or latent. Additionally, a thorough demonstration of the sorted PCMs is given. This manuscript does not address chemical thermal energy storage. Based on a thorough investigation, Francis Agyenim et al. [78] divided the PCMs into many categories according to temperature ranges: 0–65 °C for home heating and cooling, 80–120 °C for absorption cooling, and beyond 150 °C for the production of direct steam energy. Kun Du et al. thoroughly investigated [79] about categorizing PCM according to utilization temperature ranges. The latent heat storage method uses a material's transition from one phase to another in terms of its thermal characteristics within a specific temperature range. Thus, PCMs are the term used to refer to the functional materials mentioned.

Numerous studies have been done on PCM uses for thermal energy storage connected to solar heating. As is well known, the benefits of PCMs include exceptional energy storage density and outstanding dependability for thermal performance and reuse. They fit diverse applications relatively quickly because of their different phase transition temperatures and slight supercooling. In addition to the benefits listed above, PCM is an excellent choice for a thermal storage material because of its many other qualities, including its exceptional chemical stability, non-toxicity, non-corrosiveness, and malleability. The classification of PCMs and their behavior is shown in Fig. 4.

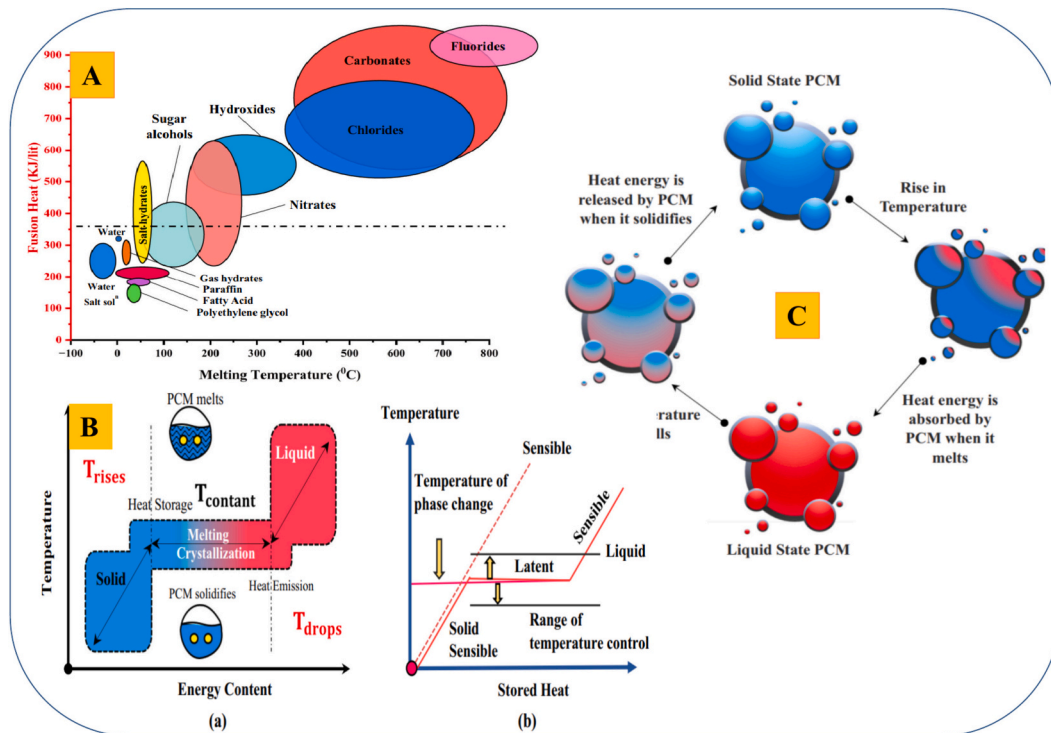


Fig. 4. A) Classification of PCM, B) Phase change behavior, C) Physical change upon cyclic heat absorption and release. Reproduced from Ref. [77]. Copyright (2024), with permission from Elsevier.

Paraffin wax is a widely favored PCM in solar heating systems due to its beneficial thermal properties and cost-effectiveness. With melting points that range between 20 °C and 60 °C, paraffin waxes are versatile and can be used in various applications, such as domestic hot water systems. Glauber's Salt, or sodium sulfate dehydrate, is another efficient PCM for thermal storage in solar systems. Its melting point is approximately 32 °C, perfect for moderate-temperature uses like space heating [80]. Additionally, fatty acids are organic PCMs that stand out for their environmental friendliness; they are biodegradable and non-toxic, and their melting points can be adjusted by changing the chain length, making them suitable for solar heating systems. Materials such as calcium chloride hexahydrate are also commonly utilized for their high thermal storage capacity and prolonged heat release capabilities, which are crucial for applications needing sustained heat delivery. Integrating PCMs into solar heating systems presents significant benefits regarding thermal energy storage and efficiency. By carefully selecting materials that meet the specific requirements of an application, these systems can efficiently capture and utilize solar energy, offering dependable and eco-friendly heating solutions [81].

4. PCM incorporation and thermal efficiency of system

PCM must be appropriately integrated into the building material to improve the building envelope's ability to store heat. The integration process must guarantee minor or nonexistent leakage during phase transition and optimize the PCM's latent heat storage capability. Researchers have devised various techniques for integrating PCM into the building envelope, as shown in Fig. 5. Direct or indirect methods can be used to incorporate the PCM into the building envelope. The most straightforward and least expensive approach is direct inclusion, in which PCM is combined directly during manufacture with building supplies, including wallboards, cement, and concrete [82]. Immersion and wet mixing were the two techniques used to add PCM directly. The PCM is immediately combined with building materials, including cement, mortar, and concrete, at the manufacturing site when using the

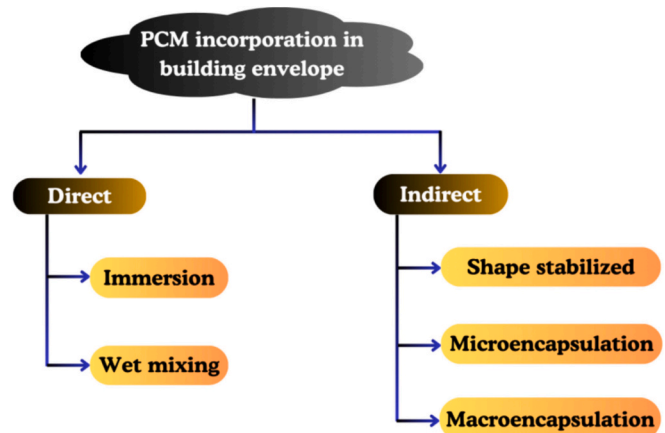


Fig. 5. Various techniques for integrating PCM into the building envelope.

wet mixing method. The thermal energy storage capacity of the building material will be improved by directly mixing PCM. Both the fact that the PCM does not impede the hydration process and that it does not react with any of the mix's ingredients are necessary conditions for the effective use of this approach. The immersion method involves dipping porous building materials (such as concrete blocks, gypsum wallboard, porous aggregate, etc.) in a container of liquid PCM. By capillary action, the building component absorbs the PCM. Temperature, PCM type, and concrete's absorption ability all affect how well and long it takes for the porous building element to soak in [83]. Several significant drawbacks were present in the construction material's two direct PCM inclusion procedures [84]. These include but are not limited to (a) PCM leakage, which may impede the hydration process; (b) PCM loss, which may result from leakage during several heating-cooling thermal cycles. (c) The immersion procedure requires a longer time—nearly 4–8 h. (d) Wet mixing might have an impact on the building element's mechanical

strength. Indirect inclusion is another way to include PCM into the building envelope. This method was first used to encapsulate the PCM before being included in the construction component. These benefits have led to the widespread usage of PCMs, enclosed in an appropriate shell material, to incorporate PCM in building components [85]. PCM is included by microencapsulation into a shell material with one mm to one-micrometer diameter. More so than 1 mm, PCM is incorporated into the shell material by microencapsulation [86]. Enhancing the PCM's thermal performance is primarily dependent on the shell material. The shell's composition must harmonize with the surrounding environment and the PCM. Various appropriate shell materials, including natural and synthetic polymers, have been proposed by Giro-Paloma et al. (2016) [87] for the fabrication of microencapsulation. Some authors have therefore investigated non-polymers, such as a shell material to encapsulate PCM, which is environmentally friendly and has high thermal conductivity, such as silica, calcium carbonate, and melamine-formaldehyde resin. Polymers are typically flammable, toxic, and have poor thermal conductivity. Due to its favorable heat transfer properties, aluminum is the most widely utilized shell material for microencapsulation [88]. Furthermore, PCM is encapsulated using copper and stainless steel [89]. Aluminum foil sheets, polymer bags, pouches, and HDPE panels are a few more shell materials that may be utilized to produce macro encapsulation [90]. PCM that has been face- or shape-stabilized is not encased in a shell material. They receive assistance from a highly porous and thermally conductive substance to lessen leakage and improve the PCM's thermal conductivity. To increase the application of PCM in the building envelope, porous materials such as clay, graphite scaffolds, polyurethane foam, and porous carbon are frequently utilized [91].

Martinopoulos et al.'s [92] analysis indicates that solar thermal systems account for at least 45 % of the overall heating demand. Its energy efficiency may be increased by improving the thermal performance, which has much promise. The two main methods that the PCM and solar thermal system may combine are thermal enhancement of the PCM. Phase segregation, heat transfer ratio, latent heat, leakage, phase change temperature, supercooling, and other parameters are PCM's main categories of dominating impact factors. There are three main types of solar thermal systems: solar collectors, thermal storage units, and solar heating terminals. The related studies will be covered in this part, along with a quick evaluation.

4.1. Thermal enhancement approaches for PCM

Enhancing the efficacy of solar heating requires the phase change material and associated components to be capable of self-behaving. Choosing PCM requires thoughtful evaluation of many variables, such as the proper phase change temperature, a large energy storage capacity, an effective heat exchanger, stable physiochemical assets, non-leakage, harmless, not corrosive inert, non-explosive, sustainable, limited phase segregation, permitted overcooling, etc. A PCM's thermal enhancement may be categorized in several ways, such as lowering super cooling/phase separation, increasing thermal conductivity, changing the phase change temperature, minimizing latent heat expansion, and many more. Initially, an illustration of PCM's improved thermal properties is shown (Table 2). In contrast to metallic PCMs, almost all pure PCMs exhibit poor thermal conductivity [93]. Because of this, the enhancement of thermal conductivity is significant for PCMs. Materials and a matrix with high thermal conductivity are two ways to increase the PCMs' thermal conductivity. Additionally, among the materials are carbon-based or metallic, such as expanded graphite, copper, silver, aluminum, and their derivatives, as well as metal oxide and nitride [94]. The mentioned thermal enhancers can improve PCM's thermal conducting performance. Furthermore, many other methods, including nanoscale encapsulation, are being explored to improve PCM thermal conductivity [95]. Using this technique, the material particles' contact area is greatly expanded, bridging them considerably closer together and producing a stronger

Table 2

TES materials and systems with varying needs that provides justifications. Reproduced from Ref. [97]. Copyright (2020), with permission from Elsevier.

	Requirement	Reason
Material		retains its original thermochemical characteristics throughout the cycling intervals
Chemical	Long-term chemical stability	
	No fire and explosion hazard	Workplace safety
Kinetic	No toxicity	Safety of operation
	No phase separation/incongruent melting	Avoiding changes on the stoichiometric composition of melt
Physical	Compatibility with storage materials and low reactivity to heat transfer fluids	Long lifetime of the container and the long lifetime of shell materials
	Small or no sub cooling	keeping the temperature constant for melting and solidification and preventing heat leakage issues
Thermal	Sufficient crystalline state	Respect for recovery system's heat transfer requirements
	High density	Reduction of the volume occupied by the TES material
Material/System	Low vapor pressure	reducing the vessel's needs for mechanical and chemical stability
	Small volume changes	Possibility of using eutectic mixtures
Thermal	Favorable phase equilibrium	Providing significant sensible heat storage
	High specific heat	TES material heat transmission is increased by minimum temperature gradients ensuring, within the parameters of operation, effective charging and discharging procedures
Material/System	High thermal conductivity in both solid and liquid states	
	Melting/solidification temperature in the desired operating temperature range	Ensuring significant latent heat storage in small volumes
Economic	Very high latent heat of transition per unit volume close to the usage temperature	supplying the TES material's complete melt and their homogeneity
	Congruent melting	
Environmental	Available abundantly	Avoiding repairs and maintenance during the TES system's lifespan
	Long lifetime	Competitive
Technological	Less expensive	
	Low manufacturing energy	lowering the systems' environmental effect and putting norms and long-term laws into place
Technological	Easy recycling and treatment	
	Low CO ₂ footprint and non-polluting	
Technological	Operational strategy	Process optimization
	Integration into the facility	Enhancing the heat transfer from the TES material to the HTF and vice versa
Technological	Efficiency	
	improvement suitable heat transfer	

heat channel, quicker heat transfer rate, and superior thermal regulating properties. Two options for enhancing thermal conducting performance in a fixed temperature differential include modifying the material's thermal conductivity and the thermal media material's contact area. Regarding the PCM's thermal conductivity, thermal augmentation techniques primarily take into account, though are not restricted to, the following factors: Expanding the region of interaction between PCM particles using techniques like nanofluid encapsulation. 2. Constructing thermal bridges between individual PCM cells; this can be done, for instance, by combining PCM with materials with high heat conductivity, such as highly conductive foam or particles. Appropriate phase change points of the PCMs are also critical to their deployment and ability to transfer heat. Different application circumstances necessitate different temperatures for the beginning phase transition. The two primary phase change point adjustment techniques are binary/ternary eutectic synthesis and selecting appropriate phase change material with a qualifying temperature. The eutectic has been shown to have the lowest phase change point of any component material, with no supercooling or phase segregation and sharp melting/freezing [96]. Its objective laws have led to the development of an equation to compute the eutectic phase change temperature. $T_e = 11 \text{ Ti Rln} x_i \text{ Hi}$ (16), where component i , x_i , R , and the gas constant (8.315 J/(Kmol)) are the component i mole fractions. T_e and T_i are the eutectic phase change points. Eq. (16) may be applied to ascertain a eutectic chemical's optimal composition mass ratio despite its binary, ternary, or even higher configuration. Both organic and inorganic eutectic PCMs are included, along with details on their thermal properties and defect suppression methods. Among the primary issues with PCMs is supercooling, separation of phases, leaks, rust, etc. The suppressive techniques will also be covered in this part, as follows. For really cooling: Overcooling is a recognized PCM downside that may be explained as phase change set off in an exothermic operation. Due to

this weakness, the material's ability to completely exploit latent heat is hindered, which results in undesired thermal energy storage and efficiency. Consequently, control of undercooling is significant for PCM applications. Many materials and techniques exist for supercooling inhibition. The subcooling defect can be reduced or eliminated using a porous matrix, salt hydrates, and other modifying materials. It may be inferred from recent research that additional treatments, such as nucleating chemicals and even appropriate system design in some situations, are acceptable in the controlled partial melting process. Regarding phase separation: Phase separation, often called phase segregation or incongruent melting, restricts the PCM's deployment ability. The phenomena mostly manifest as sediments resulting from varying compositions within the eutectic, primarily occurring during phase transition events. The PCM's lifetime is significantly reduced when an incompatible phase transition prevents it from completing another heat cycle. Regarding leakage: Because form-stable PCM is malleable and can be molded into various forms, it can be used for numerous solar heating system components. Leakage, one of its deadly disadvantages, must be removed, however. Consequently, various porous matrix, encapsulation, and material modification approaches have been developed to address this problem. Similarly, other nonmetallic materials, such as plastics and Perspex [97], have been previously explored. However, the main drawback of plastic materials for PCM-related applications is their poor heat transmission capability. The selection of materials for PCM corrosion is contingent upon the specific conditions, meaning that the choice of materials is subject to change based on various factors (Fig. 6).

4.2. Improvement of thermal properties for solar heating systems

As latent heat storage installations for solar heating have already

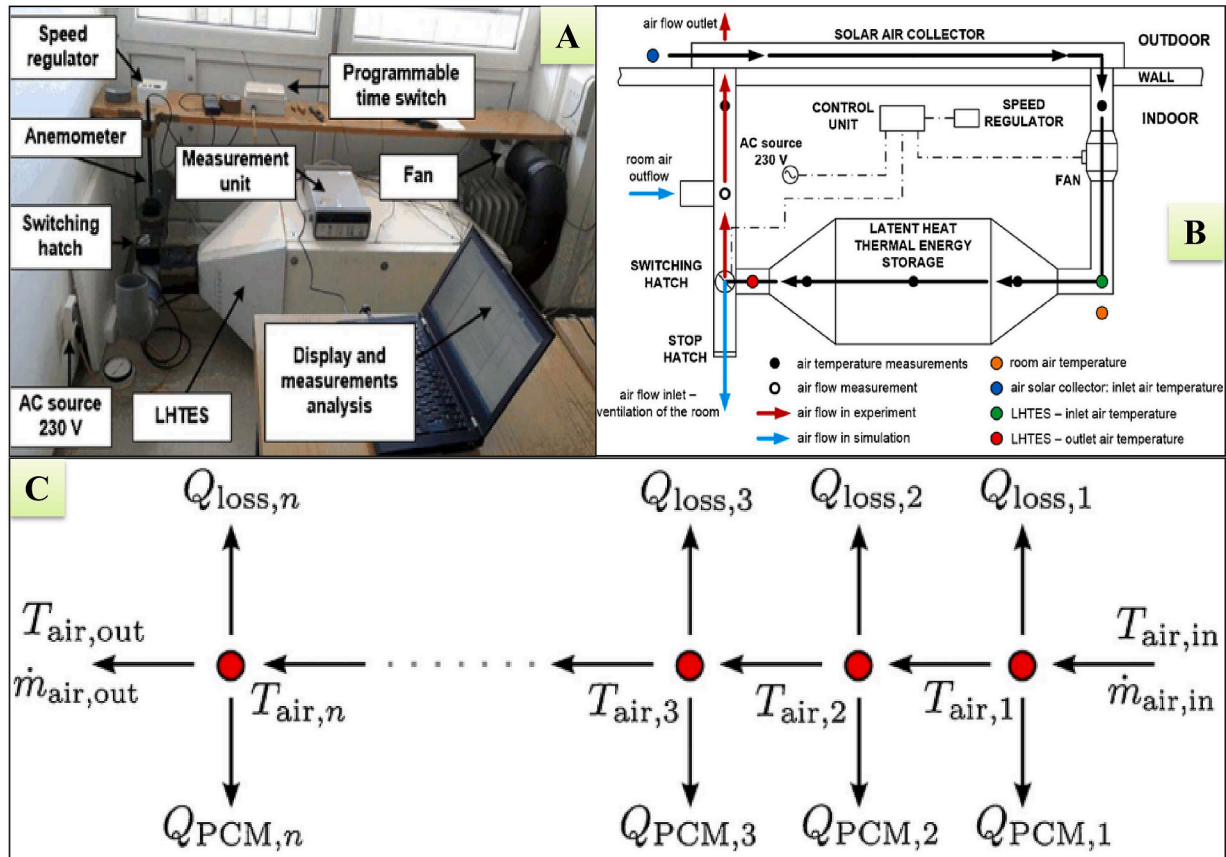


Fig. 6. A) The LHTES unit, B) Experimental setup, C) Interaction between the 1D heat transfer sub model, flowing air and surroundings. Reproduced from Ref. [96]. Copyright (2018), with permission from Elsevier.

been covered, the developments in other system components will be demonstrated here. The advancements may generally be divided into several categories. This indicates that the most remarkable improvements are structural layout, heat transfer fluid, operating strategy, annual storage cycle, energy efficiency, and so on. The references provided indicate that thermal improvements occur in several places. Initially, the structural alteration might be regarded as the most typical. A solar thermal system needs a solar collector. Fin-equipped collectors (such as longitudinal, wavy, conical, helical, planar, or serrated fins) are considered an efficient way to enhance thermal performance. Both designs showed comparable thermal efficiencies. In other words, both structures differ in how they raise the calorific efficiency. All these assessments aim to change the solar collectors' configuration to modify them thermally.

Advanced solar collector approaches, such as inserting porous filler material, polychromatic spectrum glass, booster reflectors, integration with heat pipes, and others, have been the subject of several experimental and computational studies. Configuration modifications alone won't enhance the critical components in a solar collecting system's thermal performance. Thus, the glazing technique with polychromatic spectrum has also been investigated. After trying several glazing techniques in various colors, Roozbeh Vaziri et al. [98] discovered that the collector was 35 %, 34 %, 26 %, 15 %, 11 %, 4 %, and 5 % thermally increased for the colors black, green, blue, red, violet, light yellow, and white, respectively. As a result, the glazing color change is improbable and is most likely caused by the absorbed radiation's wavelength filtering.

Furthermore, Jianjun Hu et al. [99] experimented with a solar air collector in which the heat medium rotates in active and passive modes in a swirling flow way. Additionally, it was mentioned that the thermal reinforcements had respective values of 23.83 (active) and 16.03 (passive). In addition, the current analysis has focused on the storage unit and distributing pipe network upgrades of the heat distribution system. The results of the retrofits for thermal fluid flowing creation were similarly good in terms of thermal enhancement. Graphene nanoplatelets nanofluid (0.1 wt%) and single-walled carbon nanotubes (2 vol %) were used to increase the thermal efficiency of evacuated tube solar collectors by 10 %, 35.8 %, and 4 %, respectively. Several findings exist on improving pertinent pipes in the heating distribution network's pipelines—for example, changes to the operational plan, flow pattern, and thermal insulation. When Yusuf Bas,ogul \ et al. [100] assessed the thermal insulation materials for the district heating pipeline network, they concluded that pipelines without insulation would initially cost 50 % less than those with insulation.

The heat loss was still too high when no insulating layer was used. Therefore, it is crucial that the heating pipeline be insulated. Additionally, Axel Berge et al. [101] conducted experiments on twin pipes (supply/return) covered in a vacuum-insulated polyurethane insulation material and vacuum-insulated panel cover. Basalt cloth and low-melting PET nonwovens were combined by Tingting Li et al. [102] to produce an inexpensive, lightweight pipeline thermal-insulating material. Together with strong stab/burst resistance, this material's thermal conductivity was significantly decreased to 0.0073 W/(m•K). Adding PCM on a large scale to the distribution system's pipeline insulating layer would be expensive. That's not good enough for district heating; given how much more expensive it is than conventional insulating materials. It should be emphasized that thermal energy storage units in solar heating distribution systems are the main aim of PCM deployments for district heating systems. The last chapter covered relevant research on the heating terminals of the solar heating system in such detail that more information is not required here.

5. Bio-based phase change materials

Bio-based phase change materials (BPCMs) are a broad class of materials created and widely employed in several TES systems industries to

address these problems. These materials can be classified as organic, inorganic, or composite. Agricultural waste, biomass, animal fat, plant extracts, tropical oils, other organic materials, and biopolymers are the renewable sources of BPCMs. The most successful studies on using different BPCM types in TES systems are discussed following sections.

5.1. Organic BPCMs

Sustainable biosources, such as animal fat and tropical oils like coconut, palm, and palm kernel oil, are the primary sources of biobased, non-paraffin organic PCMs. Their wide range of melting points makes them perfect for many applications. The following studies deal with organic and BPCMs that can potentially store thermal energy. Three distinct organic solvents were employed by Ong et al. [103] in 2023 to extract coffee oil from SCGs. These oils' chemistry, composition, and thermophysical characteristics were examined to assess their potential as BPCMs. According to DSC data, coffee oil extracts have a latent heat of 51.15 1.46 J/g and a phase transition temperature of around 4.5 0.72 °C. Research using gas chromatography–mass spectrometry (GC–MS) indicates that 60–80 % of coffee oil comprises fatty acids. FTIR and DSC analyses revealed that coffee oil extracts exhibited remarkable chemical and thermal stability during thermal cycling. Up to 46 min are spent keeping body temperatures below average, according to practical research assessing coffee oil extract as a possible cold therapy strategy. In their 2023 work, Ismail et al. [104] concentrated on a low-cost BPCM derived from refined vegetable oil consumables. To construct a PCM microcapsule, molten PCM was poured into holes created by chemical etching of the cenospheres. To prevent PCM from leaving the microcapsule, a silica coating was then applied to the holes. On the basis of its microstructural, thermal, chemical compatibility, and leakage characteristics, we assessed the silica-coated PCM microcapsule (SCPCM). This novel microcapsule had improved conductivity, was more thermostable, and was leak-proof. At a slower rate of around 16 °C, the PCM underwent thermal degradation due to the silica layer in SCPCM. Compressive strength efficiency increased when 10 % of the sand was replaced with SCPCM. By replacing sand with 20–30 vol% cenosphere microcapsules, an acceptable compressive strength seems to be attained. As the replacement level increased to more than 30 vol%, the ductility of the SCPCM mortar decreased due to the hydrophilic nature of the silica deposit.

Molecular dynamics simulated annealing and differential scanning calorimetry (DSC) were utilized by Phiriyawirut et al. [105] in 2023 to investigate the thermal efficiency of PCMs produced by hydroprocessing crude palm oil and using monoacylglycerols (MAGs) as an emulsifier. The start melting temperature is unaffected by emulsification, however the peak and end-set melting temperatures are raised by the addition of MAG. It's possible that the MAGs' emulsification of PCMs caused the absorption peak to expand and shift positively, increasing the operating temperature of PCMs made from palm oil. The emulsified PCMs crystallized faster, and the better MAGs interactions than alkanes (ALKs)-MAGs interactions may have contributed to the following drop in melting temperature during the second heating. This affected further emulsification processes to control the PCM properties and operating temperature. The feasibility of using biochar-based catalysts for microwave-induced catalytic biomass pyrolysis to produce gaseous fuel and enhance bio-oil in 2023 was examined by Guo et al. [106], who found that the structural characteristics and Fe-series alteration were important. The results demonstrated that the microstructure of pure biochar had an impact on its catalytic activity. During biomass pyrolysis, KOH-activated biochar with many microporous structures may greatly boost gas yields when heated with microwaves. When Fe crystal nanoparticles were added to biochar, the production of flammable gasses (CO, H₂, and CH₄) increased by 50 %, and the bio-oil became significantly simpler than when biochar was used alone. This resulted from the biochar's enhanced catalytic activity.

5.2. Inorganic BPCMs

BPCMs are made from natural resources and, like traditional paraffin-based PCMs, can absorb, store, and release large quantities of latent heat. A quantitative study of the circular latent heat thermal energy storage (LHTES) partial charging and discharging procedures was conducted in 2022 by Zadeh et al. [107]. To increase the storage's sensitivity to temperature, three aluminum foam sheets were added, separated by 120 °F. The study found that the angle (θ) and thickness (wmf) of the right-most metal foam sheet have a substantial impact on the charging and discharging processes in 16 different storage configurations. The configuration with the greatest full charging energy (case no. 7, $\theta = \pi/6$ and wmf = 7 mm) was found to not fundamentally exhibit ideal efficiency in the partial charging mode when charging and discharging according to the recommended capacity- and time-based charging/discharging energys. Rather, case number seven has up to 6 % less charging energy than the other cases when it is half charged. A Y-shaped design was found to work best when charged vertically and then rotated 60 degrees to reach maximum discharge (to a position resembling a λ). Fire-resistant phase change materials (PCMs) were developed in 2022 by Yin et al. [108] by combining ammonium polyphosphate (APP) with polyethylene glycol (PEG) loaded with poly (glycerolitaconic acid) in an approachable and ecologically acceptable manner. Thorough research was done on heat response behavior, flame retardancy, and phase change performance. The results demonstrated a significant rise in the limiting oxygen index (LOI) with the concentration of APP. The vertical testing yielded a UL-94 V0 grade, the LOI value increased from 21.6 % to 28.7 %, and the pHRR decreased by 36.15 % when the amount of APP filled reached 15 wt%. Its solvent-free method, profiles of all bio-based supporting matrices, mild curing temperature, fire safety, and high preparation effectiveness for PCM production are expected to make it a useful temperature regulator in buildings.

In 2022, Gbekou et al. [109] conducted research on micro-encapsulated phase change materials (mPCMs), which are small core-shell particles dispersed in a biobased aqueous solution. Their mechanical and thermophysical properties in cement mortars were investigated. The mortar samples were produced using a predetermined process that ensures a constant particle size and removes the potential for microcapsule leakage. Adding mPCMs significantly causes the mortar to lose its density, porosity, and mechanical strength. On the other hand, hot-disk characterization demonstrated a notable improvement in thermal performance. The system with 8 wt% mPCMs has well-balanced mechanical and thermal properties.

5.3. Composite BPCMs

Composite BPCMs reduce the negative impacts of distinct PCM classes by combining inorganic and organic components. These composites may combine the benefits of enhanced thermal conductivity from inorganic BPCMs and high latent heat from organic BPCMs. Alipour et al. [110] evaluated the thermal properties of methyl palmitate (MP) and decanoic acid (DA) binary combinations in 2023 and presented a new and renewable BPCM. Both PCMs come from biomass, the majority of which being vegetable oil. The results showed that the binary eutectic combination of 70 % DA and 30 % MP is a PCM with the necessary properties to be used as TES in construction materials. These characteristics include high values of the respective latent heats ($H_f = 202.3$ J/g, $H_m = 201.43$ J/g) and realistic melting and freezing temperatures ($T_m = 20.19$ °C, $T_f = 15.08$ °C). Analysis of its other thermophysical properties provided additional evidence that this combination met the PCM function. As the model compound, Fan et al. [111] first encased beeswax using an epoxy resin containing crystalline alkyl side chains (D18) and a number of amine curing agents. They then investigated the connection between the cross-linking network architecture and the phase change. It was demonstrated that the polarity effect and the steric hindrance impact had an affect on the phase change

performances of the polymeric FSPCMs. With M-xylylenediamine as the curing agent, the EPM-60 sample exhibited the greatest enthalpy value (146.2 J/g). Sheng et al.'s 2023 study [112] focused on cutting-edge shape-stable phase change composites (SSPCMs) for thermoelectric sensing. The matrix material was polylactic acid (PLA), and the latent heat storage media was selected to be a mixture of wood flour (WF) and PEG. The PEG/WF composites (WP) investigation showed that capillary force allowed WF to easily encapsulate PEG. When the PEG/WF/PLA ternary composites (WPCs) were combined with PLA, their high latent heat (87.2 J/g) was maintained.

6. An analysis of the use of various BPCM types in TES systems and their relative improvements

Bio-based phase change materials are gaining attention due to their potential to increase energy efficiency while having a lower environmental effect than traditional PCMs made from fossil fuels. All BPCMs are made from sustainable resources, such as biopolymers and agricultural waste. This reduces dependency on non-renewable resources and encourages sustainable energy solutions. Undoubtedly, one of the key advantages of utilizing BPCMs is this. The carbon footprint of PCMs generated from bio-based materials is less than that of their non-renewable counterparts. They assist in reducing greenhouse gas emissions and decreasing environmental deterioration. After their useful life is up, PCMs with organic bio-based foundations often biodegrade, which is good for the environment. On the other hand, some common PCMs could be challenging to get rid of. In a range of TES systems, BPCMs can improve energy efficiency by lowering temperature variations and peak energy consumption. Finally, but just as importantly, BPCMs often do not include the hazardous or poisonous ingredients that can be present in some standard PCMs. This can improve workplace security and make it simpler to follow rules. This explains the various uses of BPCMs in various industries, including waste heat recovery, construction materials, textiles, refrigeration, and heating and cooling of buildings. Despite the previously mentioned advantages, BPCMs are limited by design, as demonstrated by the thorough analysis of the included research and the review. First, because of their constrained phase transition temperature ranges, some BPCMs may only apply to a narrow temperature range in thermal energy storage devices. Furthermore, it is possible that organic BPCMs have lower thermal conductivity than their inorganic counterparts. Because the PCM can no longer transmit heat, this may lead to lower charging and discharging rates. Furthermore, the extremely high thermal conduction of inorganic PCMs may be altered by adding biomaterials. The same characteristic may cause volume variations in some inorganic BPCMs during phase transitions, resulting in material instability and structural issues with building materials or encapsulation matrices. Third, the long-term stability of BPCMs is dependent on how well they work as containment systems, heat transfer fluids, and product encapsulations. Encapsulating BPCMs properly is essential to their long-term effectiveness and prevention of leakage. Developing encapsulation techniques that are both inexpensive and durable without compromising the PCM's thermal properties might be challenging. Stated differently, inadequate compatibility might lead to decreased performance, deterioration, or leakage. Finally, from an economic perspective, producing BPCMs—especially those derived from specialized feedstock—may be more costly than producing them using standard substitutes. Maintaining cost-effectiveness while achieving scalability can also be challenging [113]. The most often employed organic BPCMs in TES systems nowadays are saturated fatty acids derived from vegetable and animal fats due to their biodegradability, sustainability, and environmental friendliness. They have a reasonable latent heat, a minor volume change during phase transition, an appropriate melting-freezing transition temperature range, and the potential to crystallize with less supercooling [113]. They still don't transfer heat as well as conventional paraffin PCMs [114]. This would account for the scientist's intention to support the melting point of saturated fatty acids

by making eutectic mixtures, or composites, of BPCMs. Regarding the thermal stability trend, the material employed affects the thermal stability of single BPCMs. For instance, organic BPCMs' thermal stability from carbohydrates, fatty acids, or plant oils varies. Certain organic materials' phase shift properties may be impacted by chemical processes that occur over several heat cycles or by degradation at high temperatures. When composite BPCMs are made of organic and inorganic components, their thermal stability depends on how well the BPCMs and the supporting elements in the composite work together. Inorganic material composites, for example, can contribute to the PCM's overall stability and durability by averting any possible degrading issues. In other words, standalone BPCMs are variable in their degree of reusability. While many BPCMs could undergo gradual degradation after repeated heating and cooling cycles, others might have a phase change temperature or latent heat capacity that varies over time. Their reusability and durability can be improved by using the proper encapsulation and confinement techniques. It's noteworthy that composite BPCMs can demonstrate superior reusability than standalone BPCMs since the supporting materials give stability. The inorganic components can strengthen their structural integrity, reducing the PCM's susceptibility to phase shifts, temperature fluctuations, and environmental degradation. In conclusion, compared to conventional organic and inorganic BPCMs, composite BPCMs can provide enhanced heat conductivity, a significant advantage. Combining inorganic and organic PCM might create strong composites with improved thermal conductivity and latent heat storage capacity. The performance of TES systems is increased overall when connected organic and inorganic components are present because they improve heat transfer during phase transitions. Composite BPCMs offer the flexibility to be aligned within the necessary temperature ranges for various applications, in addition to their ability to be produced steadily.

Consequently, the likelihood of inorganic BPCM volume changes would be decreased during phase transitions or material breakdown. Moreover, the total proportions of inorganic and organic PCMs in the composite may be changed, which is beneficial for controlling heat release and storage rates while preserving exceptional compatibility with building materials and HVAC systems. BPCMs' thermal stability and reusability may indeed be impacted by a few things, such as the encapsulation techniques and material composition. Furthermore, paraffin-based PCMs may eventually be replaced by BPCMs. The generated walnut shell carbon (WSC) and activated WSC (AWSC) were assessed for methyl palmitate (MP), which is favored as a phase transition material for thermal regulating applications in buildings, as a special form stabilizer and thermal conductivity enhancer (Fig. 7).

7. Barriers to use of bio-based PCMs

The widespread application of bio-based PCMs requires the removal of a few barriers. Since many bio-based PCMs are derived from plants and trees, they are associated with deforestation and other environmental and social challenges. Developing these materials with low environmental effects, such as farming and extraction methods, bio-fertilizers, food waste, or similar industry waste products, such as bio-based PCMs within buildings, can reduce the environmental consequences. Elevation and efficiency gains for these processes might lower the embodied carbon as raw bio-oil extraction and energy consumption have the most significant ecological effects [115,116]. Regarding energy systems and procedures, one major obstacle to be addressed is the substantial environmental effect of organic PCMs because of their high manufacturing emissions [117]. By preventing the extraction processes' contribution to embodied emissions, using recycled materials—PCM,

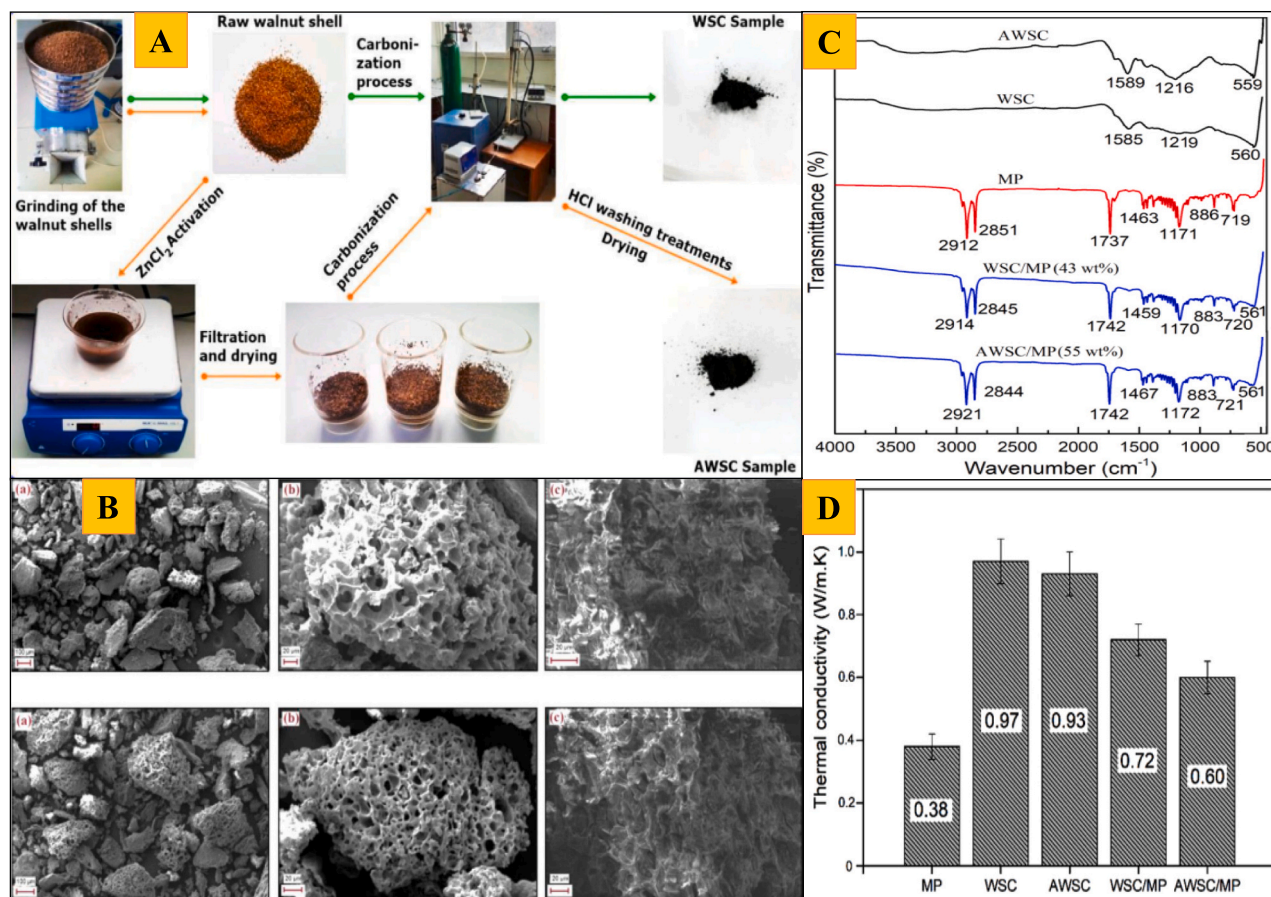


Fig. 7. A) Scheme for WSC, AWSC and WSC/MP and AWSC/MP composite PCMs. B) SEM images, C) FTIR spectra, and D) Thermal conductivity results. Reproduced from Ref. [114]. Copyright (2021), with permission from Elsevier.

carrier, or construction material—significantly reduces the environmental effect of the lifetime [118,119]. Building materials must have their environmental impact reduced through a full lifecycle review to decrease the total emissions attributable to the building stock and the overall amount of embodied carbon. The availability of materials is one potential barrier to bio-based PCMs. There might be problems with material availability if bio-based PCMs are utilized as a conditioning or thermal comfort control method. Applying different bio-based PCMs (many food-grade materials) for thermal storage creates challenges for society since there is a growing danger to food security [120]. Considering the societal challenges raised by the usage of these materials is essential. Food security and the advantages of bio-based PCMs for buildings should be evaluated jointly, not independently, as the regions from where the materials come are also recognized food sources. These concerns should be considered when conducting additional research to develop workable solutions, like the use of non-food grade PCMs or bio-based PCMs derived from food waste.

The encapsulation method that can yield the maximum proportion of PCM in the carrier material is form-stabilization [121]. This might result

in a more efficient use of bio-based PCM; however, studies have yet to be done in this field. Treating the bio-based PCM's entire system and analyzing the issues concurrently could solve the shortcomings of encapsulation and thermal conductivity (Fig. 8). This is where most current building codes come in; they look at each building element separately, not considering the structure and fire rating of the entire system. From a fire perspective, it is essential to research the fire resistance of bio-based PCM and the materials and chemicals that encapsulate them to promote their acceptability. Adopting bio-based PCMs in construction applications may be more advantageous for them, given their decreased flammability compared to standard paraffin waxes [122], provided that today's building codes are followed.

8. Final thoughts and recommendations for future

Biocomposites have garnered more interest in tandem with people's growing concerns about the status of the environment. As part of this effort, scientists are working harder to create biocomposites that are entirely biodegradable. The main advantage of utilizing ecologically

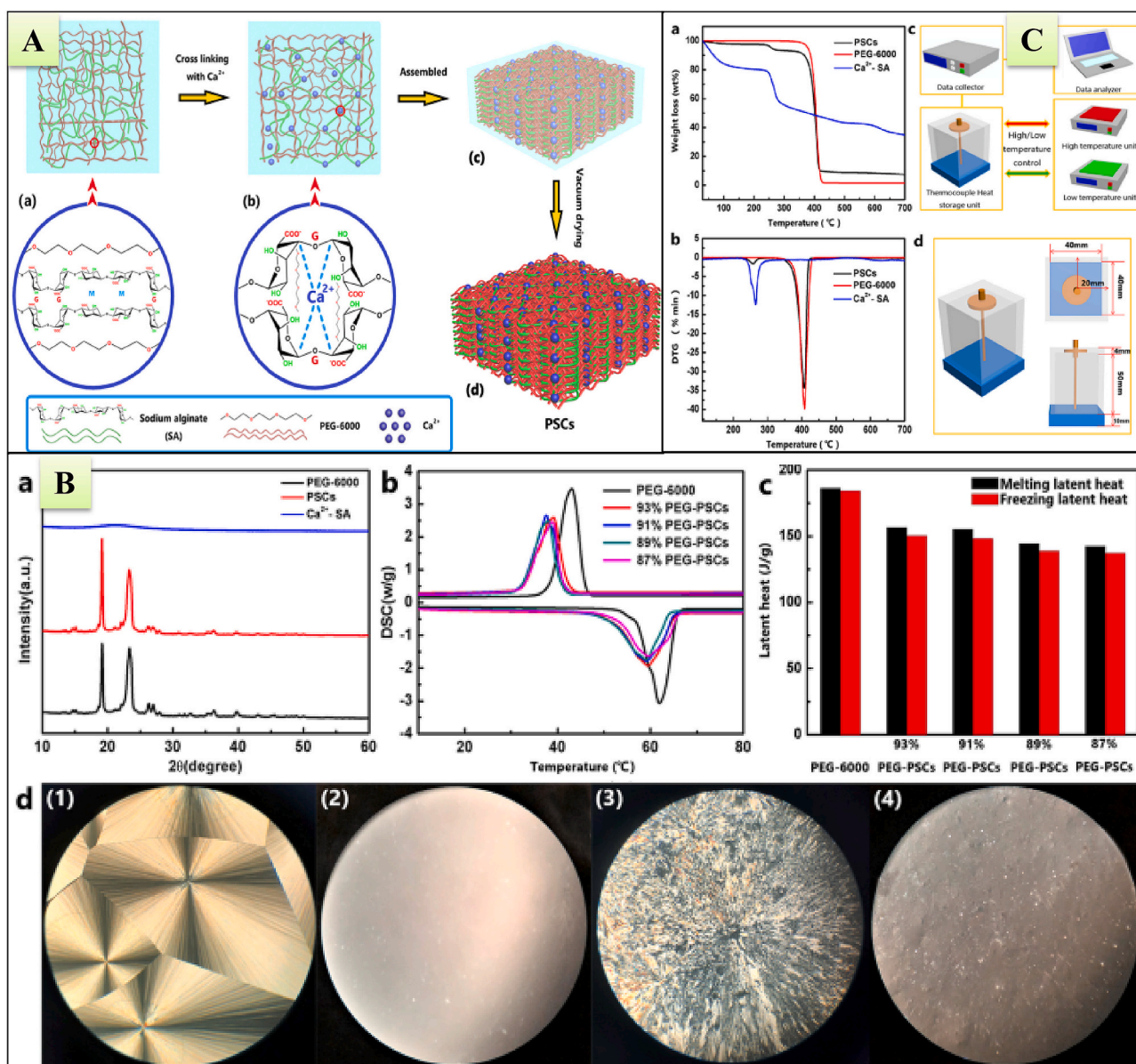


Fig. 8. A) PSCs synthesis route, B) Thermal properties with varying doping amounts, C) Thermal energy management test (TG, DTG). Reproduced from Ref. [121]. Copyright (2020), with permission from Elsevier.

friendly composites is their ability to be entirely disassembled and used. However, the current review research concentrates on synthesizing, composting, and using bio-based phase transition materials in TES and release. After reviewing more than 40 related research studies and examining the associated data, the following findings have been drawn: 1. Because BPCMs are readily available, reasonably priced, and ecologically acceptable, they may be used as excellent reinforcements to construct shape-stable PCMs. 2. The BPCMs demonstrated exceptional chemical stability and thermal reliability, qualifying them for long-term usage in many applications. 3. The rate of energy storage is accelerated by the addition of nanoparticles. 4. The melting process was accelerated, and the quantity of energy stored increased by increasing the heat provided to the BPCM. 5. The hygric inertia was abolished since there was no moisture transfer from the interior to the exterior because the PCM was positioned on the internal side. Installing PCM in the envelope center and outside has dramatically decreased the heating and cooling loads regarding thermal energy behavior. 6. Using biocompatible PCM by-products has several advantages over other biocomposites regarding thermal performance. These advantages include decreased environmental impact, economic feasibility, and acceptability for use in building applications. 7. The fire resistance duration of the LSF wall was increased to 93 min by incorporating a BPCM mat with the external gypsum plasterboard lining. 8. The application of wax combined to increase modulus and lower tensile strain simultaneously, according to the results of tensile testing. The composite fibers exhibited good thermal stability and dependability since the thermal properties did not significantly change after one hundred heating and cooling cycles. 9. The broad application of the BPCM in solar thermal energy conversion is anticipated due to its remarkable thermal stability and reusability throughout the light-to-thermal energy conversion process. 10. The poor heat conductivity of BPCMs is the primary obstacle to their widespread implementation. However, this property might be improved by using metallic carbon nanoparticles and encapsulating the material in a carbon and graphite scaffold. Notably, phase change materials are a promising and rapidly evolving technology that might help reduce total energy consumption and encourage sustainable living. This vast expanse of uncharted territory in this sector needs more organized research directed appropriately. The following suggestions aim to improve the use of BPCMs in various thermal energy storage and release applications. This will optimize the heat transfer rates through conduction and convection. 2. It is possible to research whether using exposed PCM in a suspended ceiling is feasible. This isn't a success until the acoustics and appearance of the place are enhanced. 3. It is feasible to refine the bio-based mix utilizing low embodied energy techniques, look into sub-cooling reduction, and raise the energy storage density of palm oil to produce more stable BPCMs for high-performance yet sustainable, seasonal building envelope solutions. 4. The BPCM composites containing xGnP and silica fume may be helpful when added to construction materials like mortar or concrete since they improve the material's thermal qualities. It is, therefore, necessary to consider such materials while utilizing energy-efficient construction. 5. Extensive research is required to determine the long-term endurance of BPCMs, particularly under circumstances of cyclic thermal cycling. It is essential to research degradation processes and how they affect thermal performance.

9. Summary of research

This article provides a detailed analysis of BPCMs and their role in enhancing building energy efficiency in TES systems. BPCMs are derived from natural and renewable sources and offer numerous benefits over synthetic PCMs, including sustainability, biodegradability, and non-toxicity. It emphasizes BPCMs' remarkable ability to store and release significant amounts of thermal energy during phase transitions, which makes them highly effective in regulating indoor temperatures and reducing overall energy consumption. In addition to their sustainability benefits, the article reviews the thermal and physical properties of

BPCMs, consisting of latent warmth potential, melting and freezing points, and thermal conductivity. It highlights that, although BPCMs showcase proper energy storage characteristics, there are demanding situations to overcome, including low thermal conductivity and troubles related to cloth balance. Overall, the dialogue provides a nuanced view of the ability and limitations of BPCMs in energy packages. BPCMs play a pivotal role in constructing energy efficiency in numerous applications. These progressive substances are integrated into production elements consisting of walls and insulation, heating and cooling systems, and even passive solar energy storage solutions. Their incorporation targets to noticeably lessen energy demand while enhancing thermal comfort in residential and business settings.

Despite their numerous benefits, BPCMs additionally face a few challenges. Notable advantages encompass their environmental friendliness and renewability, making them a sustainable desire for modern-day building designs. However, cloth degradation, susceptibility to oxidation, and segment segregation present hurdles that need addressing. Overall, even as BPCMs promise to beautify building energy performance, ongoing research and improvement are essential to conquer these challenges. The article outlines diverse strategies to decorate the overall performance of BPCMs. One fundamental approach involves enhancing thermal conductivity through the incorporation of nanomaterials.

Additionally, growing superior encapsulation techniques is essential for optimizing those materials. The exploration of composite materials also plays a significant role in enhancing the efficiency and effectiveness of BPCMs. The findings emphasize the long-term benefits of using bio-based solutions, especially in sustainable creation. Overall, the study affords a comprehensive evaluation of the present state of BPCMs and provides a clear direction for advancing their use in energy-efficient buildings.

10. Scope of the article

The article summarizes the modern nation of research and the applications of BPCMs in TES systems, which are specially aimed at improving building strength efficiency. In TES structures, PCMs are crucial in enhancing energy storage capabilities. Among the various sorts of PCMs, bio-based alternatives, which include vegetable oils, fatty acids, natural waxes, and animal-derived fats, stand out. Their blessings lie in their sustainability and renewability, which contribute to a more environmentally friendly method of strength storage. TES structures use those materials to seize thermal energy at some point during excessive demand, enabling it to be released while important, thereby assisting in reducing standard energy consumption in buildings. This innovative method supports energy performance and fosters a more sustainable future. A thorough examination of the thermal homes of BPCMs is famous for their latent warmth capacity, melting and freezing temperatures, and thermal conductivity. These properties are crucial for assessing their effectiveness in various applications. Additionally, it's vital to discuss their physical characteristics, including stability and the sharpness of segment transitions. The encapsulation of BPCMs also considerably complements their performance, presenting the essential support for premier capability in various environments.

Building strength efficiency can be more advantageous by integrating BPCMs into construction substances consisting of wallboards, insulation, and plaster. These materials play an essential function in passive temperature regulation within buildings, contributing to a more robust indoor climate. Additionally, BPCMs locate programs in heating and cooling systems, mainly passive solar heating and nighttime cooling strategies. Numerous case studies illustrate the energy savings and upgrades in building comfort by employing BPCMs in real-world settings. However, there are several challenges and limitations associated with their use. Fundamental limitations encompass low thermal conductivity, troubles related to lengthy-time period stability, and issues with expenses. Other demanding situations stand up from oxidation-related

issues, phase segregation, and capability leakage during section modifications, highlighting the need for energy encapsulation solutions.

Enhancement strategies play a crucial role in enhancing the performance of BPCMs. Various methods can be employed to achieve enormous advancements, such as the incorporation of conductive fillers, the manner of microencapsulation, and the development of composite formulations. Each of those techniques is ambitious in decorating thermal homes and ensuring the universal efficiency of BPCMs. Additionally, it's vital to conduct a comparative analysis among bio-primarily based PCMs and other categories of section change substances, such as organic, inorganic, and eutectic kinds. This evaluation should focus on factors like efficiency, sustainability, and cost-effectiveness, mainly in packages related to building energy performance. By comparing these elements, we can better understand the benefits and boundaries of different PCMs, leading to extra-informed decisions in choosing the most suitable substances for strength-green building solutions.

11. Recommendations for future work

Improving the thermal conductivity of bio-based phase change materials (BPCMs) is crucial for their efficiency in thermal energy storage (TES) systems. Future research should enhance hybrid BPCMs by integrating conductive materials like carbon nanomaterials, metal oxides, or graphene. This integration must maintain sustainability and biodegradability while boosting thermal performance.

To ensure long-term effectiveness, key research areas include addressing oxidation, phase separation, and thermal cycling degradation. Optimizing BPCM encapsulation with biodegradable polymers or protective coatings can enhance stability against environmental effects.

A significant challenge is the high cost of BPCMs compared to traditional materials like paraffin. To promote wider adoption, cost-effective manufacturing methods should be explored, including using waste materials like animal fats or vegetable oils. Optimizing synthesis and processing technologies and conducting lifecycle cost analysis will help assess commercial viability in large-scale applications.

Exploring new bio-based phase change materials (BPCMs) presents significant opportunities, mainly through biopolymers, algae-derived substances, and agricultural by-products. This research aims to enhance PCMs' thermal properties and sustainability by utilizing underexplored natural resources. Improving encapsulation techniques is crucial to prevent leakage and material degradation, and innovations like micro- and nano-encapsulation should be further investigated.

Integrating BPCMs into building systems and energy storage through smart thermal energy systems is essential for optimizing energy performance. This involves real-time data to manage energy storage and release efficiently. Additionally, examining the interaction between BPCMs and smart grids or IoT technologies could enhance energy savings.

Analyzing the environmental and lifecycle impacts of BPCMs is vital for assessing their sustainability as energy storage options. This includes evaluating their carbon footprint, energy consumption during production, recyclability, and end-of-life disposal. Further research should compare BPCMs with synthetic materials and conduct pilot studies in diverse environments to validate their effectiveness and document real-world energy savings.

BPCMs extend beyond thermal energy storage by integrating various building components to enhance energy efficiency, moisture regulation, and acoustic insulation. Research shows significant potential for bio-based PCMs in improving building energy efficiency. However, addressing limitations like thermal performance, long-term stability, cost, and system integration is crucial. Future studies should connect lab findings to real-world applications through innovative technologies and environmental assessments. With ongoing interdisciplinary efforts, BPCMs could be vital in advancing sustainable building technologies.

12. Conclusion

This overview highlights the growing importance of PCM in TES systems as a sustainable solution for improving building energy efficiency. BPCMs, sourced from renewable and biodegradable materials, offer several benefits, including minimal environmental impact, non-toxicity, and significant latent heat storage capacity. When incorporated into building materials, BPCMs have the potential to reduce energy consumption and enhance thermal comfort. They contribute to energy-efficient designs through passive temperature regulation and integration with solar energy, thereby reducing reliance on traditional resources and promoting sustainability. However, the evaluation identifies challenges such as the low thermal conductivity of BPCMs, long-term stability issues, and higher costs compared to synthetic PCMs. Addressing these challenges requires research focused on improving thermal performance through encapsulation, enhancing stability, and reducing costs via green manufacturing practices. Future studies should target critical areas that can advance BPCM technology. Investigating innovative composites further improves their performance. Field trials and large-scale demonstrations are essential to validate their effectiveness. Additionally, developing innovative TES systems with integrated controls will optimize energy management. Finally, conducting comprehensive life-cycle assessments will emphasize the environmental benefits of BPCMs. While BPCMs show promise for thermal energy storage in buildings, further research is needed to fully understand their potential for sustainable and energy-efficient construction. Overall, this analysis underscores the significance of BPCMs in contributing to energy-efficient building designs and supports the transition toward more sustainable TES solutions.

CRedit authorship contribution statement

Shweta J. Malode: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Investigation, Formal analysis, Conceptualization. **Nagaraj P. Shetti:** Writing – review & editing, Visualization, Validation, Resources, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

This is a review article.

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